

Periodontology 1

Course Overview and Introduction

ODON 621

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Assistant Professor



Mission: To advance the science and art of health through
education, service, scholarship and social accountability

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1

In this presentation, we will discuss



Course Description

1. Assess and diagnose periodontal health and disease
2. Comprehensive treatment planning process
3. Prognosis of the dentition
4. Prevention of disease
5. Provide periodontal therapy
6. Recognize oral mucosal and osseous disorders (*Oral Path/Oral Medicine*)
7. Identify severe/complex periodontal disease
8. Refer the patient to a specialist



Course (Learning) Objectives

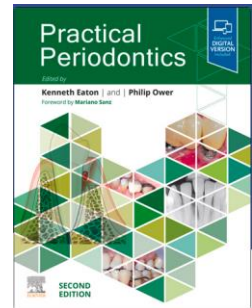
- 1. Perform a complete periodontal examination & document findings**
- 2. Recognize extraoral and intraoral signs and symptoms of periodontal disease**
3. Teach effective patient-specific plaque control procedures/skills
4. Perform initial therapy consisting of supragingival removal of plaque, calculus, extrinsic stains and a re-evaluation
- 5. Identify clinical signs of disease and compose the data collected to reach a periodontal diagnosis**
6. Evaluate the progress of patient during treatment and appropriately recommend referral to specialist

Perio 1 Grading (*Not Staging and Grading...*)

In-Class Assessment/Quiz =QUIZZES		20
Group Assignment/Small Group Activity=CASE STUDIES AND WORKSHEETS		20
Course Project		
Reflection		
Mid-Term Examination=M/C 40 questions		25
Final Examination=		
Objective Structured Clinical Exam (OSCE)=Case Based in Examsoft		30
Clinical Requirements		
Entrustable Professional Activities (EPA)=show up, on time and participate		5
Skill Assessments		
Other (describe)		
Total		100%

"Practicalities": Success

1. Canvas announcements to communicate weekly
2. If you need something: email Jennifer.goss@cnsu.edu
3. Assigned prereading: listed in Canvas (Practical Periodontics, articles)
4. Quizzes: weekly, at start of class, previous week's material. If you are late=can not take
5. Upload Cases/Assignments Daily: work in groups (A-C') to discuss and present, **each person uploads individually**. Late 50% credit, or less
6. EPA's: Not there, no points. Don't participate, late, unprofessional or not engaged...lose points



Absence Request Form (excused)

Email Absence Request Form: follow instructions

Prior to sending the Absence Request Form, *please complete the following blanks on the form:*

1. Session, Dates Missed, Session Faculty
2. Deadline to Complete Make-up Plan: 1 week from your absence date
3. Make-Up Plan: Review the materials for the missed session. Ask a classmate for notes. Make an appointment for office hours if you have any questions regarding the material.

If there was an Assignment from that day, you have 1 week to complete it and email it Dr. Goss (please check first if the assignment is still open in Canvas, if so please upload it there).

(Note: Quizzes cannot be made up, EPA points cannot be made up).

Once you have completed the make-up (if there is one), resend completed form and I will sign it. Then forward it to Candace Fox and Sara Treanor.

Resources

- American Academy of Periodontology (AAP): charts and papers
 - Post on Canvas, www.perio.org "Research and Science" tab
- European Federation of Periodontology (EFP):
 - www.efp.org "Publications" tab
- Carranza and Newman text-if need to further your knowledge

Course Outline

- 1) Understand the disease process of periodontal diseases and conditions:

Anatomy ,Pathogenesis, Biofilm, Host Response,
Periodontal medicine, Risk assessment, Risk factors

- 2) Assessment of periodontal condition: Clinical parameters, Classification, Diagnosis, Prognosis

- 3) Treatment planning and referral

- 4) Periodontal treatment:

Instrumentation (WILL OCCUR IN “PERIODONTAL INSTRUMENTATION course), Mechanical debridement, Patient education, Adjunctive treatment

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10

Course Outline-1st 8 weeks

Anatomy and Pathogenesis

Host Response and Biofilms

Periodontal Assessment: Clinical Parameters

Classification: Periodontal Health and Disease

Diagnosis

Periodontal Medicine (Medications, Systemic Disease)

Risk Assessment & Risk Factors

Course Outline-2nd 8 weeks

Clinical Examination and Treatment Planning

Prognosis

Patient Education

Adjunctive Treatment

(Mechanical Debridement: occur in instrumentation)

Re-evaluation, Referral, Advanced Treatment Options

Clinical Scenarios







What is periodontal disease?



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16

Periodontitis Definition

- chronic
- multifactorial
- inflammatory disease
- associated with dysbiotic plaque biofilms
- progressive destruction of the tooth-supporting apparatus:
 - loss of periodontal tissue support
 - clinical attachment loss (CAL)
 - radiographically assessed alveolar bone loss
 - presence of periodontal pocketing
 - Presence of gingival bleeding

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17

AAP 2017

Periodontal Diseases/Conditions

- What do you see in the photo?
- What causes it?
- What does it do to teeth/mouth/patient?
- How do we identify and diagnose it?
- How do we explain it to patients?
- How do we treat it?



Teeth are important....



Significance of Periodontitis

- Major public health problem due to its high prevalence (46 % US adults >30 years old have clinical attachment loss)
- Leads to tooth loss and disability
- Negatively affects chewing function and aesthetics, be a source of social inequality, and impair quality of life
- Accounts for a substantial proportion of edentulism and masticatory dysfunction, results in significant dental care costs
- Plausible negative impact on general health ("systemic link")

Periodontal disease

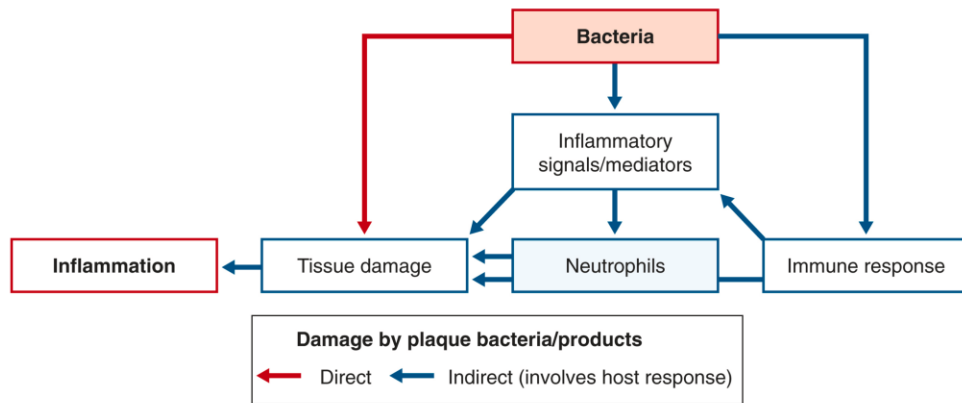


FIGURE 1.2.4 Direct and indirect bacterial effects in periodontitis.

Eaton p. 14

What is happening here? Pathogenesis of tissue damage.

Inflammation

- complex response to harmful stimuli typically from a pathogen, trauma, or toxins (or autoimmune)
- **blood vessels dilate** and become more permeable, and attract more **immune cells and fluid** into local tissue
- classical signs of inflammation are heat, pain, redness, swelling and can lead to a loss of function
- can be acute or chronic

Periodontal Treatment



Bacteria



Host



Interaction → Result

We have bacteria and host....and a result. How can we modify the bacteria and or the host? You will see several examples of modifying the bacteria...as well as some of the physical properties of the host. How can we modify the systemic factors of host? Can we directly effect immune response?



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What are periodontal diseases and conditions?

- World Workshop 2017 AAP
- Classification-definition



Classification at-a-Glance



2018 Classification of Periodontal and Peri-Implant Diseases and Conditions

Periodontal Health, Gingival Diseases and Conditions

- Periodontal Health and Gingival Health
- Gingivitis: Dental Biofilm-Induced
- Gingival Diseases: Non-Dental Biofilm-Induced

Periodontitis

- Necrotizing Periodontal Diseases
- Periodontitis
- Periodontitis as a Manifestation of Systemic Disease
- Periodontal Abscesses and Endodontic-Periodontal Lesions

Periodontal Manifestations of Systemic Diseases and Developmental and Acquired Conditions

- Systemic Diseases or Conditions Affecting Periodontal Supporting Tissues
- Mucogingival Deformities and Conditions
- Traumatic Occlusal Forces
- Tooth- and Prosthesis-Related Factors

Peri-Implant Diseases and Conditions

- Peri-Implant Health
- Peri-Implant Mucositis
- Peri-Implantitis
- Peri-Implant Soft and Hard Tissue Deficiencies

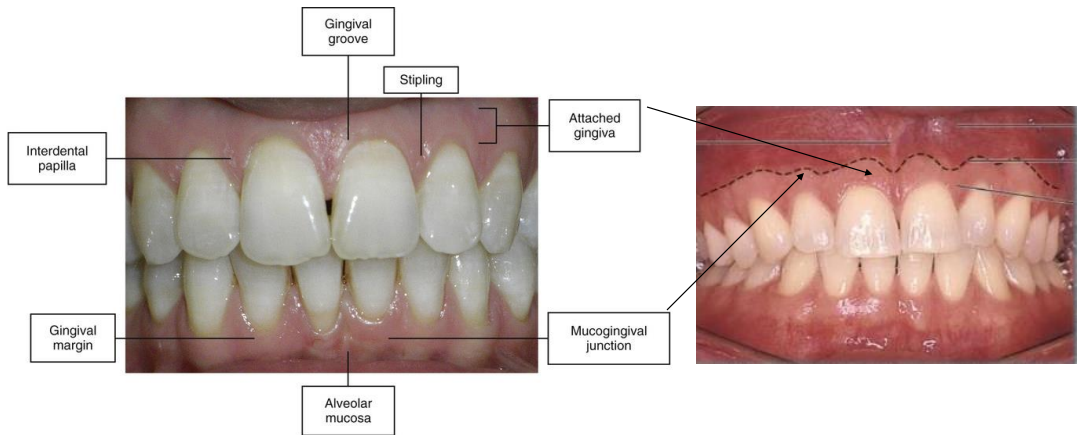
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Health vs. Disease



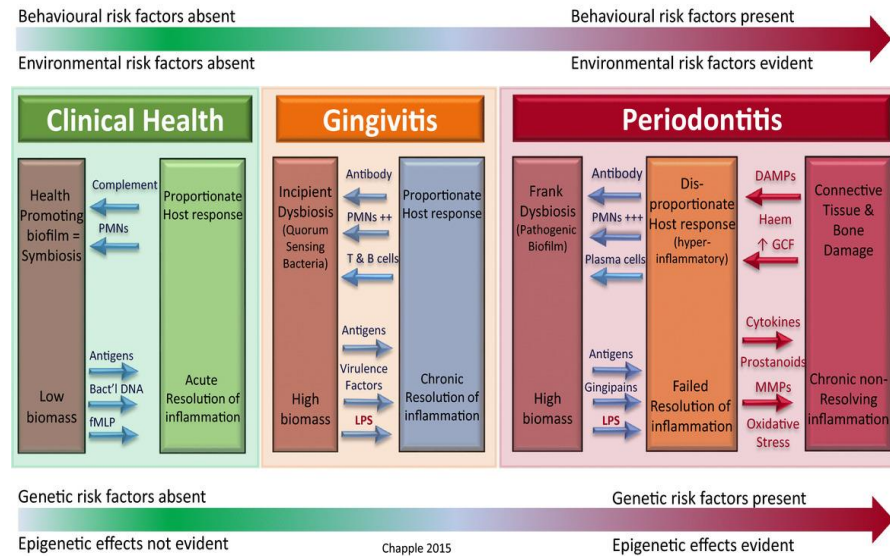
Anatomy



Loss/Changes Anatomy



Pathogenesis: How a disease develops

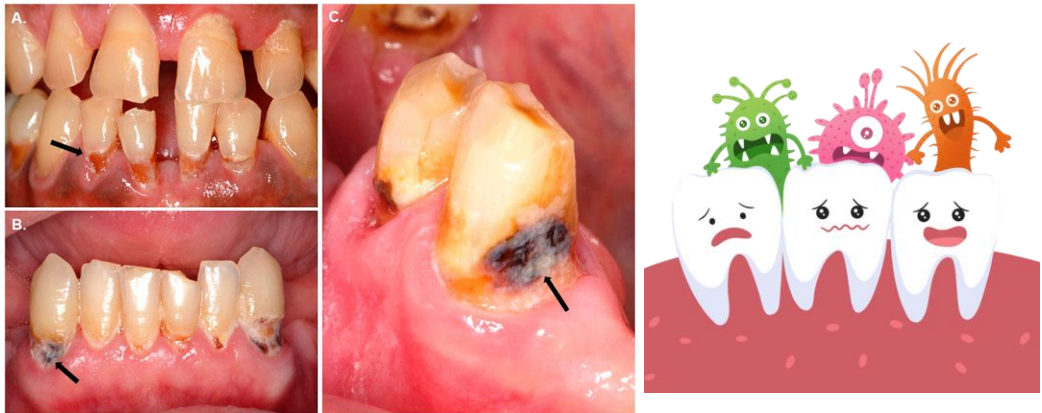


Journal of Periodontology, Volume: 89, Issue: S1, Pages: S17-S27, First published: 21 June 2018, DOI: 10.1002/JPER.17-0095

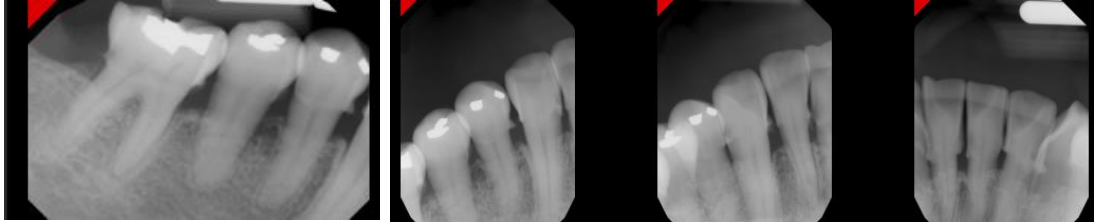
Contemporary model of host–microbe interactions in the pathogenesis of periodontitis, in which the host response drives an incipient dysbiosis (gingivitis). If the biofilm is not disrupted/removed, frank dysbiosis results and perpetuates a chronic nonresolving and destructive inflammation. DAMPs, damage-associated molecular patterns; fMLP, N- formylmethionyl- leucyl- phenylalanine; GCF, gingival crevicular fluid; LPS, lipopolysaccharide; MMPs, matrix

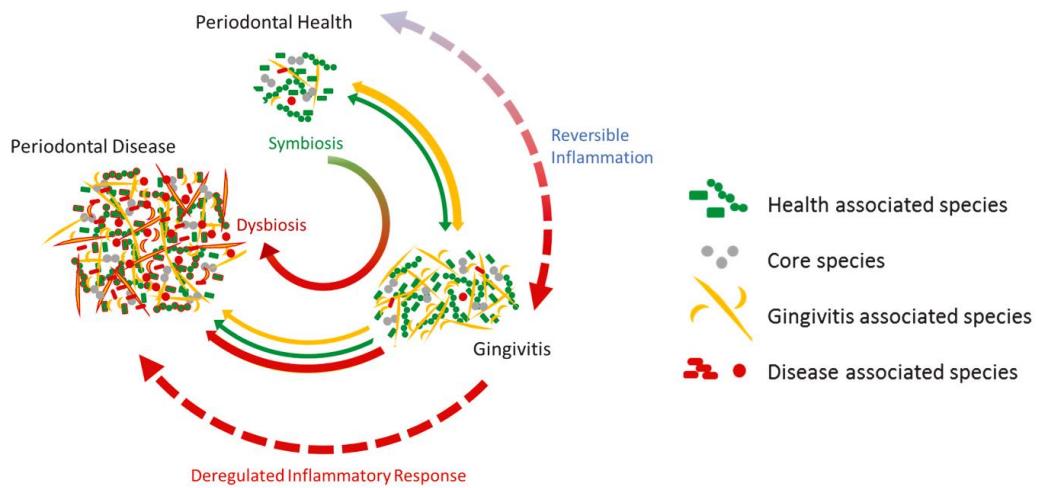
ARTICLE, AUTHOR, AND PUBLISHER.

Biofilm

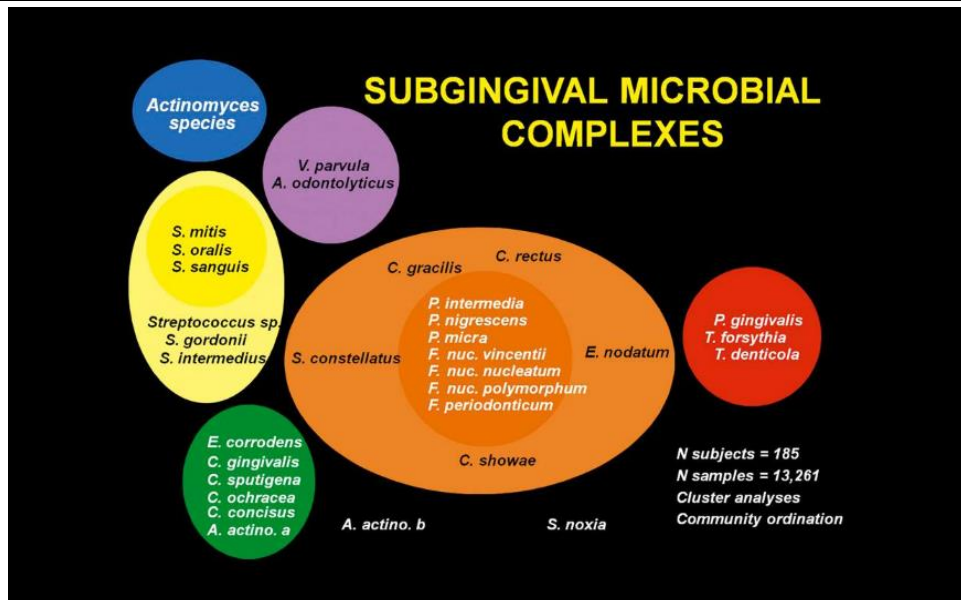


Biofilm ↔ Host response



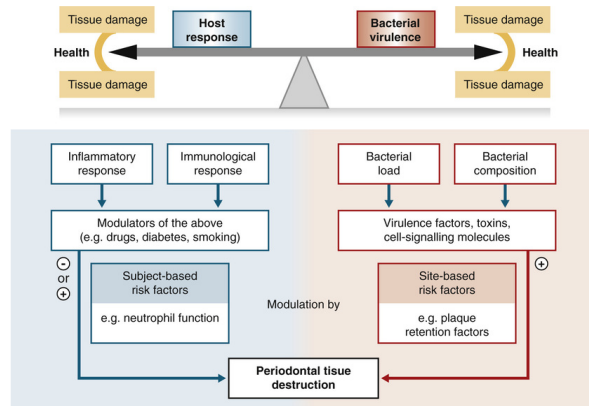


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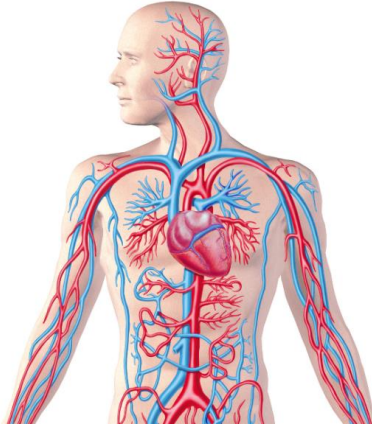


California Northstate University College of Dental Medicine Socransky, et al; Carranza text

Host Response and Susceptibility



Systemic conditions associated with periodontal disease

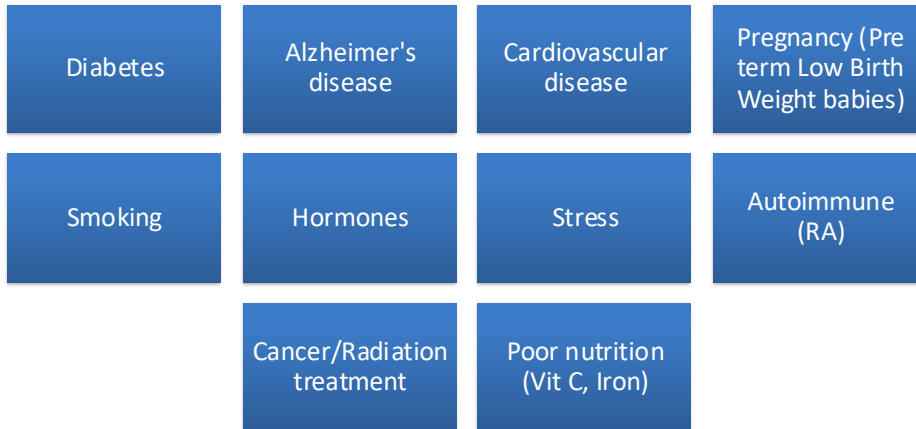


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38

There is an important two-way relationship between systemic health and periodontal disease.

Systemic conditions associated with periodontal disease



What is a risk factor?

Risk factors are something that increases the chance of developing a disease. They are modifiable and can include lifestyle changes.

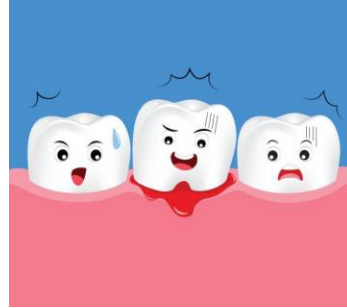
Examples for Periodontal Disease:

- Tobacco usage (Smoking and Smokeless)
- Diabetes
- Stress
- Medication(s)
- Poor nutrition
- Genetics and Family History
- Age
- Gastric Esophageal Reflux Disease (GERD)(GRD) and Bulimia
- Lack of Regular Dental Care
- Rheumatoid Arthritis

What is a risk factor and how does it relate to periodontal disease? A risk factor is something that increases the chance of developing a disease. It has to be biologically plausible to cause disease and has been shown to precede the development of disease. It does not necessarily mean that a patient with these risk factors will definitely get the disease. On the other hand, absence of a risk factor does not mean that they won't get the disease. They are modifiable and can change also, like improving lifestyle factors. Here are common examples of risk factors for periodontal disease, smoking, diabetes, stress, medication, poor nutrition, age, GERD, lack of regular dental care and rheumatoid arthritis. Patients with rheumatoid arthritis have limited manual dexterity and cannot brush as well.

Assessment of periodontal condition:

1. Symptoms of patient
2. Medical and Dental History
3. Visual Assessment
4. Periodontal Examination
5. Radiographic Assessment
6. Assessing local etiologies



Assessment: Periodontal Examination

1. Probing depths
2. Plaque index
3. Bleeding on probing/exudate
4. Recession
5. **Clinical attachment level**
6. Tooth mobility
7. Furcation involvement
8. Fremitus
9. Occlusion



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42

The most important part of assessment when it comes to treatment plan, diagnosis is CAL (clinical attachment level/loss)

What is a treatment plan?



A detailed plan with information about the patient's disease, goal of treatment, treatment options, possible risks, prognosis and expected length of treatment.

Periodontal Treatment Plan Phasing

- Based on Diagnosis, Etiology, Prognosis
- Periodontal Treatment Plan Phases:
 - Initial Phase (Phase 1 Periodontal Treatment)
 - Non-surgical Phase
 - Second Phase (Phase 2 Periodontal Treatment)
 - Surgical Phase
 - Third Phase (Phase 3 Periodontal Treatment)
 - Maintenance Phase
 - Restorative Phase



Regardless of diagnosis, the treatment plan will be divided into phases. Maybe different than restorative Phasing (or ODTP/Axium Phasing)

Phase 1: Initial Treatment

- Periodontal Exam
- Oral Hygiene Instructions
- Smoking cessation
- Mechanical debridement
- Extraction of “hopeless” teeth
- **One Month Re-Evaluation**



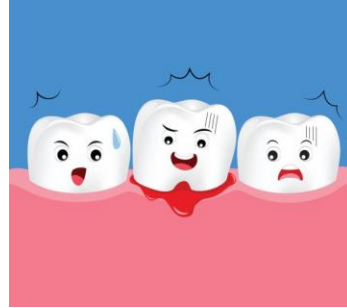
Memorize this. It's important.

Phase 2: Surgical Phase=Perio 2 (Spring)



Assessment of periodontal condition:

1. Symptoms of patient
2. Medical and Dental History
3. Visual Assessment
4. Periodontal Examination
5. Radiographic Assessment
6. Assessing local etiologies



Cases

- Visual Assessment:

- Gingival description: color, contour, tone, texture 
inflammation, tissue loss/edema  disease process???
- Presence of plaque/calculus/stain
- Condition of teeth: decay, incisal wear, malocclusion
- Condition of oral cavity (xerostomia?)

Cases

- Work within your Group to answer questions on cases
- You may share a document amongst your group, or have individual document
- We will review cases together (one group will present case)- edit answers if needed based on discussion
- Each **person** will upload case writeups into Canvas assignment by 5 pm day of assignment

Radiographic Interpretation "Reps"

Questions?

